

Zifan Li

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EDUCATION

University of Alberta

Bachelor of Science in Computing Science (Specialization)

Edmonton, AB

Expected 2027

Relevant Coursework: Software Engineering, Machine Learning, Database Management, Web Applications, Reinforcement Learning, Visual Recognition, Search and Planning in AI, Human-Computer Interaction

TECHNICAL SKILLS

Languages: Python, Java, C, SQL, JavaScript, TypeScript

Backend/Data: FastAPI, Django, REST APIs, PostgreSQL, SQLAlchemy, Alembic, pgvector, Docker, Git, Next.js, Streamlit

AI/ML: PyTorch, RAG, Embeddings, Vector Search, LLM Evaluation (LLM-as-judge, golden datasets), LangGraph, LangChain, LLM APIs (Anthropic, OpenAI, OpenRouter), Structured Output, Prompt Engineering

EXPERIENCE

Waiter

New Club Cafe

May 2024 – Sep 2024

Vulcan, AB

- Orchestrated service operations in a high-volume environment, prioritizing tasks to maintain efficiency and customer satisfaction standards
- Facilitated seamless communication between front-of-house and back-of-house teams to resolve service bottlenecks and ensure timely delivery
- Managed client relationships and conflict resolution, adapting quickly to changing requirements in a fast-paced setting

PROJECTS

AI Content Generation Pipeline | *Python, PostgreSQL/pgvector, SQLAlchemy, LangGraph, LLM APIs, Docker* Apr 2026 – Present

- Designed a multi-stage LLM generation pipeline (five code-gated stages plus a LangGraph tool-calling research agent) joined by strict Pydantic contracts, where each stage composes its output from the upstream artifact's typed fields instead of trusting free-text model output; verified by a 700-test suite that fakes every paid external API
- Built a config-driven provider layer routing 17 LLM call sites across 4 models through one YAML registry, with contract-identical structured-output wrappers over two vendor APIs, so swapping a stage's model is a one-line config change; unknown seats and missing credentials fail loudly instead of silently defaulting
- Enforced cost governance before spend: bounded the research agent with per-call and cumulative budget ceilings plus a no-new-results halt condition, and added a preflight that prices every paid job from measured rates, with a dry-run mode and a resume manifest making reruns idempotent
- Implemented deterministic validate–repair–escalate guardrails on LLM output (structural and budget contract checks, exactly one bounded repair, then a typed hard failure) and a two-layer verification gate pairing citation-grounding against fetched sources with a cross-model-family judge, catching contract violations before any paid call
- Built the retrieval and evaluation layers: BGE-M3 embeddings in a PostgreSQL/pgvector schema of 14 tables evolved over 21 Alembic migrations, queried through an HNSW cosine index and fed by a structural-first chunker that keeps facts from splitting across chunk boundaries; paired with an eval harness persisting each metric run keyed by git SHA and dataset version for a queryable regression timeline

AI Portfolio Assistant | *Python, FastAPI, LangChain, ChromaDB, JavaScript*

July 2025 – Jan 2026

- Built an end-to-end RAG chatbot integrated into a personal portfolio website to answer queries about me
- Implemented a document ingestion pipeline to chunk and embed profile data into ChromaDB
- Developed backend services and APIs with FastAPI to handle context retrieval and generation via OpenAI/LangChain
- Designed a user-facing chat interface with JavaScript for seamless interaction with the AI backend

Heart Disease Predictor | *Python, XGBoost, Streamlit, SHAP*

July 2025 – Aug 2025

- Engineered a predictive system by benchmarking Logistic Regression, Random Forest, and XGBoost models, achieving 88% accuracy with the selected XGBoost classifier
- Optimized model performance using GridSearchCV for hyperparameter tuning, achieving improved generalization
- Implemented SHAP visualizations to interpret model predictions and identify key risk factors
- Deployed an interactive web interface allowing users to input health metrics and receive real-time predictions

CERTIFICATIONS

Associate AI Engineer for Developers | *DataCamp*

Nov 10, 2025

Machine Learning Specialization | *Stanford Online*

June 27, 2025